Nature, Scope and Methodology of Macroeconomics

- Micro Vs Macro
- Methods

"I'm a social scientist, Michael. That means I can't explain electricity or anything like that, but if you ever want to know about people I'm your man."

Nature and Methodology

- **Economics –** divided into 2 main sub-fields of Micro and Macroeconomics
- **Microeconomics** is the study of how individual consumers or households and firms make decisions and how they interact in specific markets for goods and services
  - It looks at individual markets—for corn, records, books, and so forth—that operate within the broad national economy
- Focuses on human behavior of wealth creation at analytically disaggregated levels
- Key questions:

Nature and Methodology

- What determines the price of particular goods and services?
- What determines the output of particular firms and industries?
- What determines the wages workers receive? The interest rates lenders receive?
- The profits businesses receive?
  - How do government policies—such as minimum wage laws, price controls, tariffs, and excise taxes—affect the price and output levels of individual markets?
  - Why do incentives matter inside firms and how can economic theory be used to properly structure a firm's incentives to increase worker productivity and firm profitability?

Nature and Methodology

- **Macroeconomics** the study of economy-wide phenomena, that is human economic behaviours in terms of the whole economy and its interaction with other economies
  - It is concerned with the behaviour of the economy as a whole, with booms and recessions, the economy's total output of goods and services and the growth of output, the rates of inflation and unemployment among others:
- Typical macroeconomic questions include:
  - *What determines the general price level? The rate of inflation?*
  - *What determines national income and production levels?*
  - *What determines national employment and unemployment levels?*

Intro

- *What effects do government monetary and budgetary policies have on the general price, income, production, employment, and unemployment levels?*

Nature and Methodology

- **Methodology** –refers to the way in which economists go about the study of their subject matter. There are two broad approaches :
- ***Positive Statement*** –this investigates how different economic agents in society seek to improve their welfare using wealth.
  - Positive economics is that branch of economic inquiry that is concerned with the world as it is rather than as it should be
  - It is very factual and is concerned with what is, what was, what will be (as in how the world actually works)
  - It is objective and can be replicated by others or to proved wrong by empirical evidence.
  - Positive statements are testable and any disagreements over them are appropriately handled by an appeal to facts

Nature and Methodology

- **Normative economics** is that branch of economic inquiry that deals with value judgments—with what prices, production levels, incomes, and government policies ***ought* to be**
  - It involves suggesting **what should be** or ***ought to be*** the situation or outcome
  - It involves making value judgements and sometimes taking strong moral positions – inextricably bound up with philosophical, cultural and religious positions
  - It can involve professional subjectivity but are never testable

Methodology: Nature of positive economics

- **The Economist as a Scientist/Positive economist**
  - Involves thinking analytically and objectively.
  - Requires that something that is positive and testable should emerge from theories somewhere –for if it does not, their theories will be unrelated to the real world.
  - Scientists seek to answer positive questions by relating them to evidence
  - In general. Theories are used in explaining **observed phenomena** . A successful theory enables us predict the consequences of various outcomes
  - Observation concern consequences of events. Any explanation whatsoever of how these events are linked together is a theoretical construction

Methodology: Nature of positive economics

- Theories are used to impose order on the observation, to explain how what is seen is linked together.
- Theory is like economic models are a simplified representation of a particular feature of the world that we want to understand.
  - Theory and models are used to simplify reality in order to improve our understanding of the world
  - Theories are expected to have the following elements:

i. A set of definitions that clearly define the variables to be used. A variable is some magnitude that can take on different possible values. Variables need to be carefully defined. Price is an example of an economic variable. Price is the amount of money that must be given up to purchase one unit of that commodity.

Methodology

ii. Set of assumptions – outlining the conditions under which the theory apply. Assumptions have to be realistic . The art in scientific thinking is deciding which assumptions are more realistic to make. Assumptions aid at simplifying the model. Different assumptions will be made for different models Includes developing abstract models from theories and the analysis of the models

- Example the PPF we looked above we assumed two products
- Using the ***ceteris paribus,*** or ***all else being equal,*** *assumption*, economists study the relationship between variables while the values of other variables are held unchanged or constant.
- The ceteris paribus device is part of the process of abstraction used to focus only on key relationships

Methodology

iii. **Hypotheses** – about the relationships of variables . Hypothesis is statement about how two or more variables are related to each other: For example –quantity produced depends among other things, upon price **ceteris paribus**

iv. **Prediction /Outcomes** – a prediction is a conditional statement that takes the form: *if you do this , then such and such will follow. The* predictions of the model follow logically from the assumptions and hypothesis. The outcome maybe refer to the equilibrium of the model.

- ***Overall, a model is a formal statement of a theory. Models are descriptions of the relationship between two or more variables. They can be graphs mathematical etc.***

- The relationships among the above factors can be written as follows:
  - $D_c = f(p_C, Y, P_s, P_s, P_f, Z)$
- This models simply says that –the demand (Dc) for the car depends on or is dependent on or is a function of the explanatory variables listed in the brackets.
- The models give us the framework to organize how we think about a problem, data is required to quantify the observed relationship in our theoretical model
- A theory or model that has been tested several times and proved true over time is called a behavioral law.

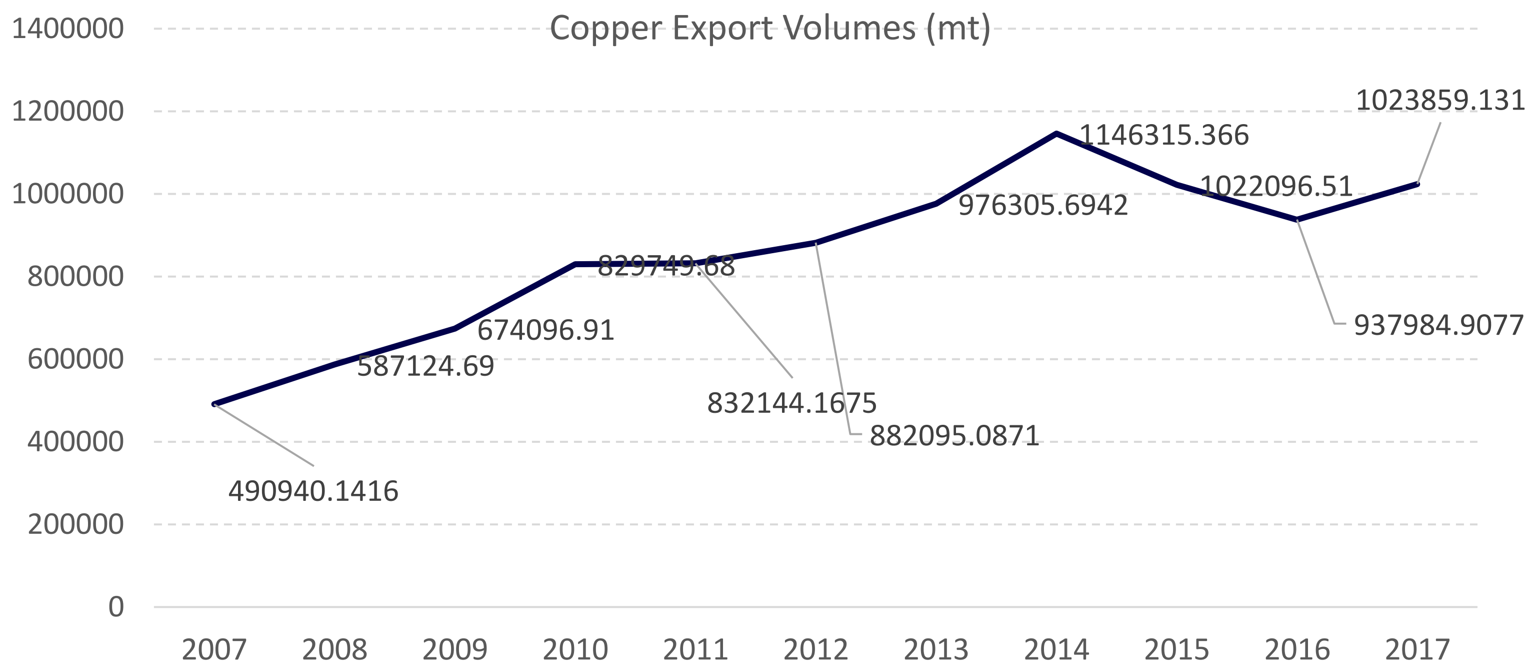
Methodology

- So a ***behavioral law*** is a sensible theoretical relationship not rejected by evidence over a long period
- To test the model, we need to have information or data of the variables.
  - ***Economic data*** refer to pieces of evidence about economic behavior –example total amount of cars sold per month in the country, the average monthly price of new cars and used cars, the average price of fuel etc. It can refer to total output,
  - Data can be presented in many different ways. These include:

a. ***Time series data -*** *this is a sequence taken at successive equally spaced points in time. For example, we can collect the price of the same book like Begg et al. every month for the whole year. The series of prices make what we call time series data*

Methodology

- Example of time series data: the Zambia statistics Agency compiles the volumes of copper every year. This data is presented in the figure below.



Methodology

b) ***Cross sectional data*** - is a type of **data** collected by observing many subjects (such as individuals, firms, countries, or regions) at the one point or period of time,
   - For example, we can ask each one of our class members present today to give us their gender, phone ownership are (if they have a phone or not), income (how rich they) etc.
   - This information is only collected for today and we do not collect it again. We have get what we call a cross-sectional data

c) ***Panel Data*** – this is a mix of time series and cross section. For example, we can get the information in b) above every month from the same people. We make a panel.

Methodology

- **Economic Experiments**
  - Because economics is a science which is centred on human behaviour, it is not always possible to conduct experiments the way natural sciences like physics or biology do
  - But some experimentation can still be undertaken
- First, experimentation can be conducted in a “laboratory” where data can be collected through:
  - observations on individual or group behaviour
  - Questionnaires and surveys or analysis of existing or available data such as wages, prices etc.
  - Example: Thaler –conducted a number of experiments to explore how individuals respond when faced with different questions on losses and gains in relation to a reference point

- For example, Thaler found that prior ownership of a good -e.g., a football game ticket, changes people's willingness to sell, even at a higher price than that which they paid. He called this an endowment effect, and it has been replicated in many studies
- Second natural experiments – where the phenomenon is determined by the natural conditions which are not in the control of the experimenter
  - Natural experiments can be exploited when some change occur which allows observation to be carried out on the effects of this change in one population and comparison made with another group who are not affected .
- Third experimental economics – is a branch of economics that studies how to design and implement experiments to study particular economic questions

- Example: Testing the effectiveness of a new drug. The company can select a group of individuals willing to participate in the trial and would provide incentives
- The participants are randomly divided into two groups:
- One group is given the drug - referred to as the treatment group
- and another group (the control group) is given a placebo (something resembling the real drug but has no chemical effects)
- After some time, the two groups are compared and any differences in outcomes may be attributed to the drug.
- The individuals are assigned to either groups randomly (that is by chance and not by choice). Hence the experiments are called randomized control trials

Methodology

- **Diagrams, line etc**–are vastly used to show relations among variables.
- They are often drawn on assumption that other variables remain unchanged "*ceteris paribus*"
- *Graphs can depict relationship between two variables*
  - *For example the relationship between the amount your dad earns and the amount he saves*
  - *If you dad makes a lot of money, he is likely to save more*
  - *We will have a positive relationship between the two*
- *Graphs can also depict some condition that is achieved when two variables are combined. Example PPF?*